

TrustFile Complete Reviewer From OOP to Dependencies

PART 1: Object-Oriented Programming (OOP)

Object-Oriented Programming (OOP) is a programming paradigm that organizes code using classes and objects.

Pillar	Definition	Example in System
Encapsulation	Bundling data and methods	User class with attributes
Inheritance	Child inherits from parent	User(db.Model, UserMixin)
Polymorphism	Same method different behavior	db.session.add()
Abstraction	Hiding complexity	db.session.commit()

Your system is OOP because it uses classes, objects, inheritance, and encapsulation.

PART 2: Attributes

Attributes are variables that belong to a class (like database columns). Total: 27

Class	Attributes	Count
User	id, username, email, password	4
UploadedFile	filename, filepath, hash, risk_score, etc.	13
VTCache	file_hash, positives, scan_date, etc.	5
UploadFileForm	file, submit	2

PART 3: Variables

Variables store data values. Types include local, global, parameters, and class attributes.

Estimated total variables in your system: ~95

PART 4: Parameters vs Arguments

Parameter: variable in function definition.

Argument: actual value passed when calling a function.

Type	Example
Parameter	def scan(file_id):
Argument	scan(60)

PART 5: Modules vs Dependencies

Type	Definition	Example
Module	Your Python file	malware_api.py
Dependency	External package	flask, requests
Built-in	Included in Python	os, hashlib

PART 6: Dependencies Used

Dependency	Purpose
flask	Web framework
flask_sqlalchemy	Database ORM
flask_login	User authentication
flask_wtf	Forms
wtforms	Form fields
werkzeug	Security
requests	API calls
mysql.connector	Database connection
openai	AI analysis

PART 7: Quick Summary

Category	Count
Classes	4
Attributes	27
Variables	~95
Parameters	29
Functions	~28
Modules	2
Dependencies	9

PART 8: Sample Questions

1. Is your system OOP? Yes, because it uses classes and inheritance.
2. Parameter vs Argument? Parameter is in definition, argument is actual value.
3. Dependencies? 9 external libraries installed via pip.

PART 9: Glossary

Term	Definition
Class	Blueprint for objects
Object	Instance of class
Attribute	Variable inside class
Parameter	Variable in function
Argument	Value passed
Module	.py file
Dependency	External package